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**Macro-Micro Robotic System with a
Flexible Continuum Tip for Minimally
Invasive Surgery**

Interim Report and Project Plan

Abstract

This draft literature review is structured for a thesis on a UR5e-based macro-micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work in the *Lachlan Report*, identifies which foundations remain valid, and highlights where the state of the art now requires updated evidence. The review prioritizes three technical threads central to this thesis: (1) tendon-driven continuum manipulator design and motion algorithms, (2) integration with a macro robot (UR5e) with improved teleoperation behaviour, and (3) rapid end-effector exchange mechanisms. It also includes explicit “missing area” headings to direct next-stage searching, reading, and synthesis.

ORIGINALITY STATEMENT

‘I hereby declare that this submission is my own work and to the best of my knowledge it contains no materials previously published or written by another person, or substantial proportions of material which have been accepted for the award of any other degree or diploma at UNSW or any other educational institution, except where due acknowledgment is made in the thesis. Any contribution made to the research by others, with whom I have worked at UNSW or elsewhere, is explicitly acknowledged in the thesis. I also declare that the intellectual content of this thesis is the product of my own work, except to the extent that assistance from others in the project’s design and conception or in style, presentation and linguistic expression is acknowledged.’

Signed: _____

Date: _____

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Nomenclature

(The nomenclature must respect the following order: latin symbols, Greek symbols, acronyms. Within each category of symbols, follow alphabetic order. An example is provided below)

A	=	amplitude of ...
C_p	=	...
C_x	=	...
C_y	=	...
D	=	...
α	=	...
TLA	=	...

I Introduction

I.1 Project Context and Problem Significance

This thesis extends prior macro-micro robotic surgery work by coupling a UR5e macro manipulator with a new flexible continuum tip, then improving movement algorithms inherited from earlier implementations. The applied significance is strong: MIS demands dexterity in constrained spaces, low tissue interaction force, high positional repeatability, and robust safety controls.

From the earlier project, the macro-micro concept and teleoperation feasibility are already established at a proof-of-concept level. The key opportunity now is to move from demonstration toward a more clinically credible engineering argument through stronger modeling, improved control reliability, and modular hardware architecture (rapid tool exchange).

I.2 Working Thesis Gap Statement (Draft)

Current literature and prior project work support feasibility of macro-micro MIS manipulation, but leave a practical translation gap at the intersection of **accurate continuum movement control, stable and reliable teleoperation, and fast, repeatable end-effector reconfiguration**. This thesis addresses that gap on a UR5e-based platform by improving inherited motion algorithms and integrating reliability and modularity mechanisms necessary for repeatable experimental deployment.

II Literature Review

II.1 Introduction to Minimally Invasive Surgery (MIS) and Robotics

- The clinical motivation: Why do we need surgical robots? (Precision, dexterity, patient outcomes).
- The rise of Macro-Micro systems (combining a large positioning arm like a UR5e with a delicate continuum tip).

Historical progression of surgery and MIS robotics Technology and surgery have been inseparable for centuries as every surgical advancement or care improvement has been enabled by the parallel development of instruments and tools. From anaesthesia to antiseptics, innovation has enabled surgeons to reduce risk and expand the scope of treatable conditions (Emerging Robotic Platforms for Minimally Invasive Surgery) In the 19th century, innovations such as the transition from candlelight to electric lighting allowed for practical internal visualisation, marking the beginning of a major paradigm shift

in the surgical field (Hargest five thousand years). Electricity also enable the development of various specialised surgical procedures, such as electrocautery for haemostatisis and the use of electric drills for orthopaedic surgery (hargest 5000 years). These innovations allowed for a stronger focus on precision and dexterity, and techniques such as laparoscopy and endoscopy were developed at the beginning of the 20th century, resulting in some of the first minimally invasive surgeries (MIS)(Hatzinger M, Fesenko A, Sohn M. The first human laparoscopy and NOTES operation: Dimitrij Oscarovic Ott (1855-1929). Urol Int. 2014;92(4):387-91. doi: 10.1159/000358016. Epub 2014 May 20. PMID: 24852454.). This trend towards less invasive procedures has continued, with the development of revolutionary robotic surgical systems such as the da Vinci Surgical System in the late 1990s, which has become widely adopted for various surgical procedures (Intuitive Surgical. da Vinci Surgical System. <https://www.intuitive.com/en-us/products-and-services/da-vinci/surgical-systems>. Accessed June 2024).

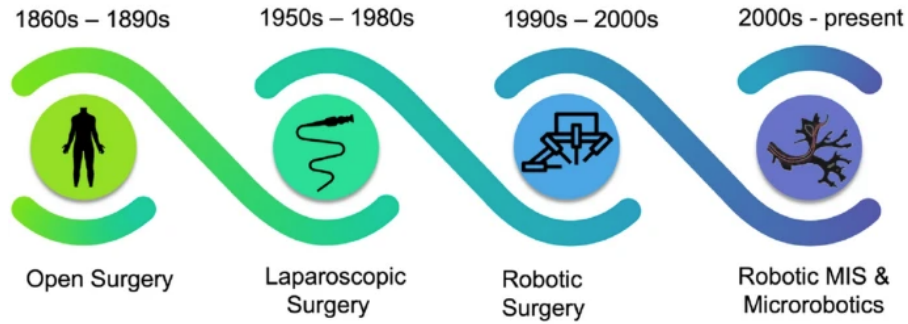


Figure 1: Progression of surgery and enabling technology from conventional approaches to modern MIS robotics.

Macro-micro architecture motivation The modern frontier of surgical robotics is achieving human-level dexterity, accuracy, and force control in increasingly small and complex anatomical spaces. One key technological direction enabling this transition is minimally invasive surgical tooling with a macro-micro architecture. The macro unit provides large-workspace positioning and efficiency; however, its larger structure can be prone to deformation and vibration. The micro manipulation unit is then attached to the larger arm and provides the distal dexterity required for manipulation in confined anatomy, where surgical targets often require end-effectors in the 3–4 mm class for practical access through narrow channels and natural orifices. In current snake-robot literature, this micro stage is typically tendon/cable-actuated to achieve high local curvature and fine orientation control, but its performance depends strongly on motion transmission quality, sensing, and control architecture (Seetohul and Shafiee, 2022). In a macro-micro arrangement, this local dexterity complements the macro arm’s gross positioning capability and helps isolate the endpoint from macro-stage vibration through coordinated control (Scott, 2022).

Cost and accessibility context With growing demand and a rapidly expanding surgical robotics market, there is ongoing debate over whether the high cost of integrated platforms such as da Vinci limits accessibility, with acquisition costs reported up to \$4 million AUD and per-procedure costs of roughly \$1000–\$5000 (Do the costs of robotic surgery present an insurmountable obstacle? A narrative review). Despite clear clinical benefits, providers must still evaluate cost-effectiveness, especially in resource-limited settings (Do the costs of robotic surgery present an insurmountable obstacle? A narrative review). Evidence from low-to-middle income settings further shows that adoption barriers are not limited to purchase price alone: recurring maintenance, disposable instrument costs, and sustained training requirements can prevent programme continuity even after initial installation (Raza et al., 2024). The same review also highlights system-level constraints, including limited infrastructure, workforce training gaps, and tension between investing in robotics versus addressing unmet basic surgical needs, reinforcing a clear translational gap for scalable low-cost platforms (Raza et al., 2024). Taken together, these findings indicate a specific gap: affordable macro-micro systems that combine low-cost hardware with delay-robust, clinically credible teleoperation performance are still underdeveloped. These findings support the broader literature position that low-cost macro-micro architectures, including off-the-shelf hardware and open-source software stacks, are an important direction for improving accessibility while maintaining MIS precision and safety requirements (Lachlan’s report).

UR5e baseline and observed limitation Specifically, the UR5e is a widely used, cost-effective robotic arm that can serve as the macro stage, with reported pricing around \$50,000–\$60,000 AUD, and can be integrated with a range of custom micro systems (Universal Robots price guide: What to expect (new and used costs)). The use of a UR5e-based macro-micro architecture has been demonstrated in prior thesis work by L. Scott (2022), which provided a proof-of-concept teleoperation pipeline using open-source ROS2. However, as highlighted by Scott, translational macro-arm commands produced observable vibration in the integrated system. Reducing this behaviour is necessary to meet clinical expectations for safety, repeatability, and motion quality in operating environments.

Continuum tips are a clinically relevant class of hyper-redundant end-effectors for MIS because they combine high dexterity with slender geometries that can be advanced through confined access pathways (e.g., lumens, natural orifices, and small incisions), where conventional rigid instruments are mechanically constrained. In the tendon-driven subclass, curvature is generated through differential tendon tension about a central backbone or articulated serial structure, enabling multi-DOF distal steering with compact proximal actuation and making these devices well suited to macro-micro architectures (Scott, 2022; Razjigaev et al., 2021; Seetohul and Shafiee, 2022).

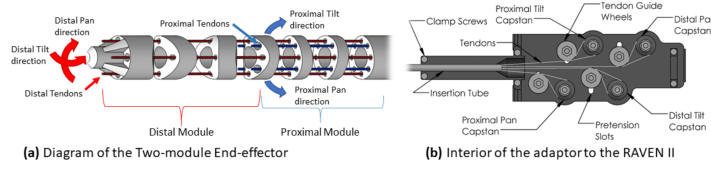


Figure 2: SnakeRaven macro-micro surgical manipulator concept used as a relevant continuum-tip reference from prior work (Scott, 2022; Razjigaev et al., 2021).

Within this context, current UR5e-based continuum implementations in the local literature follow the tendon-driven snake-like micro-manipulator direction established in Lachlan Scott’s thesis, including a 4 mm-class distal tool, macro-micro integration with a UR5e platform, and teleoperation via a haptic interface. Scott reports preliminary electromagnetic-tracking validation showing close pose mirroring under teleoperation (reported average absolute position errors of 0.0272 mm and 0.0161 mm across trial groups), while also identifying practical engineering constraints that remain unresolved, including macro-stage translational vibration and the time-consuming exchange of tendon-driven end-effectors in the current modular workflow (Scott, 2022; Razjigaev et al., 2021; Razjigaev et al., 2022; Seetohul and Shafiee, 2022).

II.2 Teleoperation and Control in Surgical Robotics

- *The Problem:* Human tremor, system latency, and mapping user intent to rigid vs. continuum kinematics.
- *Current Solutions:* Review papers on teleoperation filters (Kalman filters, moving average, predictive kinematics).
- *The Gap:* Discuss how high-end systems do this well, but open-source/low-cost systems (like Lachlan’s) suffer from vibratory feedback across ROS/external control nodes.

Within this control context, the relevant evidence can be read in five linked layers: tremor suppression, latency compensation, robust state estimation, predictive bilateral control, and intent-aware shared autonomy.

Human physiological tremor is a central control challenge in robot-assisted MIS because involuntary operator motion is transmitted through the teleoperation loop and can appear as vibration at the surgical instrument tip, degrading precision during fine tissue manipulation, suturing, and needle handling. Sang et al. (2016) report that tremor suppression methods must balance two competing requirements: strong attenuation of involuntary motion and minimal phase delay, as latency in the command pathway can directly reduce teleoperation quality. Their zero-phase adaptive fuzzy Kalman filter (ZPAFKF)

combines a zero-phase prefilter with adaptive Kalman estimation to improve tremor estimation and compensation in real time, and experimental master-slave results show lower error and reduced tip vibration relative to standard filtering baselines. This is directly relevant to the present thesis because it frames tremor control not as a standalone signal-processing problem, but as a coupled teleoperation and tracking problem in which compensation quality, delay, and manipulator dynamics must be considered together (Sang et al., 2016).

System latency is a second critical bottleneck in bilateral teleoperation, particularly when haptic feedback is included. Batty et al. (2022) show that communication delay creates a fundamental transparency-stability trade-off: increasing direct force correspondence can degrade stability, while aggressive stabilisation can mute the fidelity of operator feedback. In response, they implement an environment-estimation and force-prediction architecture, where slave-side contact parameters are estimated online (EWRLS) and master feedback is generated through a predicted virtual environment rather than delayed direct force reflection. Their human-in-the-loop palpation experiments demonstrate that this predictive approach substantially reduces force prediction error (with the Hunt-Crossley model outperforming Kelvin-Voigt and direct force feedback), and improves teleoperation transparency under delay. A key implication in the literature is that latency management should be treated as a control-architecture problem, not solely a communications problem, and that model quality strongly influences the practical quality of teleoperated manipulation (Batty et al., 2022).

Filtering is therefore not optional in surgical teleoperation; it is a foundational requirement for maintaining stable and trustworthy state estimates when measurements are noisy and network transport is imperfect. Lashari et al. (2025) show that a Kalman filter paired with MOESP-identified state-space models can estimate patient-side manipulator position with high robustness under delay, jitter, and packet loss, while remaining computationally lightweight for real-time use. This is especially relevant when exact plant dynamics are not available a priori, because MOESP builds the model directly from surgical kinematic data (JIGSAWS), allowing filtering performance to be grounded in observed system behaviour rather than idealised assumptions. In practical terms, filters are needed to smooth stochastic disturbances, reconstruct missing or delayed information, and preserve control continuity so that downstream motion control and haptic/visual feedback remain clinically usable. A key limitation, however, is that the reported performance is primarily based on dataset-driven modelling and simulated network impairments, so in vivo generalisability and cross-platform transfer still require further validation (Lashari et al., 2025).

Predictive bilateral control is also emerging as a practical strategy for continuum and snake-like teleoperation under delay. Ebrahimian et al. (2024) propose an online architecture that combines model-mediated force prediction on the master side with ANN-based

operator motion prediction on the slave side, with the explicit goal of reducing perceived communication lag while preserving position synchronisation and haptic transparency. This is highly relevant to the macro-micro literature because it addresses the same core bottleneck as latency and filtering studies, but at the controller-architecture level: delay compensation is handled through coordinated forward and backward prediction, rather than through one-sided compensation only. Their simulation results show strong delay bypass for known constant delays in snake-like robot teleoperation, but a key limitation is that performance is demonstrated mainly in simulation with structured assumptions (e.g., known constant delay and constrained robot DOFs), which may not fully capture variable-delay clinical networks or higher-complexity surgical interaction dynamics (Ebrahimian et al., 2024).

Beyond tremor and delay, user intent mapping is a central teleoperation design problem in shared control. Bowman et al. (2024) argue that direct motion mapping alone is insufficient for robust telemanipulation when the human and robot hands are physically dissimilar, because low-level geometric tracking does not necessarily preserve high-level task success. Their intent-based task-oriented shared control framework instead blends operator-following behaviour with autonomous task assistance using inferred intent probabilities and adaptive arbitration weights. Critically, this contribution is not just another blending policy: it formalises how authority should shift based on hand-structure discrepancy and task constraints, which is directly relevant to macro-micro MIS systems where distal tools cannot perfectly mimic surgeon hand kinematics. This supports the broader literature position that intuitive teleoperation should be evaluated by both task completion and perceived control, rather than endpoint tracking error alone (Bowman et al., 2024).

However, an important limitation is that the reported validation is constrained to controlled grasping tasks and specific intent models, so transferability to highly dynamic, deformable, and safety-critical in vivo surgical contexts remains an open question.

Collectively, these control studies establish clear performance requirements for low-cost macro-micro platforms and provide a direct conceptual bridge to modular continuum hardware discussions.

II.3 Modularity and Tendon-Driven Continuum Manipulators

- *The Problem:* Continuum robots are excellent for navigating complex anatomies, but they are usually monolithic. If a tool breaks or a different tip is needed, the whole robot is swapped.
- *Current Solutions:* Review mechanical coupling techniques in robotics (magnetic tool changers, mechanical twist-locks).

- *The Gap:* Analyze the specific challenge of maintaining tendon tension across a boundary. Why hasn't a quick-release mechanism been widely adopted for micro-continuum robots yet?

II.4 System Safety and Reliability Considerations

- *Brief Review:* Safe velocity limits, force-aware operation, and fault handling in MIS workflows.
- *Context:* How these constraints pair with motion smoothing to improve repeatability and operator confidence.

II.5 Next-Step Research Flow and Predicted Subheadings

1) Continuum tip modelling and control limits *Focus:* tendon friction, hysteresis, backlash, coupling, calibration drift.

Why next: this is usually the biggest source of endpoint error in macro-micro MIS setups.

Look for: model-based vs data-driven correction; real-time feasibility.

- **Predicted subheading A:** Physical sources of nonlinearity in tendon-driven continuum tips.
- **Predicted subheading B:** Modelling approaches for friction, hysteresis, and backlash.
- **Predicted subheading C:** Identification and calibration workflows under drift.
- **Predicted subheading D:** Real-time control feasibility and computational constraints.

2) Macro-micro error compensation strategies *Focus:* how macro vibration/deformation is compensated at the micro/end effector.

Why next: directly bridges your intro into your control contribution.

Look for: feedforward/feedback split, PD vs nonlinear/robust control, stability assumptions.

- **Predicted subheading A:** Error propagation from macro stage to distal tip.
- **Predicted subheading B:** Feedforward and feedback compensation architectures.
- **Predicted subheading C:** PD, nonlinear, and robust control comparisons.
- **Predicted subheading D:** Stability assumptions and validation conditions.

3) Collision detection for flexible distal tools (safety-oriented framing) *Focus:* continuum body collision checking, not just rigid-link approximations.

Why next: this is likely a core thesis gap and safety argument.

Look for: geometric methods, signed distance fields, real-time performance limits.

- **Predicted subheading A:** Geometry representations for flexible tool bodies.
- **Predicted subheading B:** Real-time collision checking under control-rate constraints.
- **Predicted subheading C:** Safety envelopes, warning layers, and bounded response logic.
- **Predicted subheading D:** Integration with teleoperation and shared-control safety workflows.

4) Safety-constrained control (clinical relevance layer) *Focus:* virtual fixtures, active constraints, force/velocity limits, safe-stop behaviour.

Why next: lets you justify why your method is more than “better tracking.”

Look for: enforceable constraints during teleop/autonomy-assisted motion.

- **Predicted subheading A:** Virtual fixtures and active constraints for MIS.
- **Predicted subheading B:** Force and velocity limiting strategies.
- **Predicted subheading C:** Safe-stop and fault-handling behaviour.
- **Predicted subheading D:** Constraint enforcement during teleoperation and shared control.

5) Teleoperation signal conditioning and motion smoothing *Focus:* tremor filtering, predictive smoothing, delay compensation, human-in-the-loop stability.

Why next: cleanly ties back to the practical usability problems in prior work.

Look for: quantitative trade-off between smoothness and responsiveness.

- **Predicted subheading A:** Signal conditioning methods for surgical teleoperation.
- **Predicted subheading B:** Predictive smoothing and delay compensation.
- **Predicted subheading C:** Human-in-the-loop stability analysis.
- **Predicted subheading D:** Performance trade-offs: responsiveness vs smoothness.

6) Modular end-effector/tool exchange mechanisms *Focus:* quick-change coupling while preserving tendon routing/tension and repeatable calibration.

Why next: strong practical novelty angle and likely under-addressed in literature.

Look for: repeatability metrics after swap, re-zeroing time, failure modes.

- **Predicted subheading A:** Mechanical coupling strategies for quick exchange.
- **Predicted subheading B:** Tendon routing continuity and tension retention after swap.
- **Predicted subheading C:** Post-swap calibration and re-zeroing procedures.
- **Predicted subheading D:** Reliability metrics and failure-mode analysis.

II.6 Synthesis of Literature and Knowledge Gaps

Across recent literature, there is strong agreement that high-quality surgical teleoperation depends on coordinated management of tremor, delay, state-estimation uncertainty, and operator intent. However, most validated architectures are demonstrated either on high-cost integrated systems or under simplifying assumptions (e.g., known constant delay, controlled tasks, or simulation-only evaluation), which limits generalisability to resource-constrained and clinically variable settings.

In parallel, continuum and snake-like end-effectors provide clear distal dexterity advantages for confined MIS workspaces, but persistent control difficulties remain around tendon friction, hysteresis, backlash, calibration drift, and repeatability after hardware reconfiguration. Evidence from low-to-middle income contexts also shows that long-term adoption depends not only on capital cost but on maintenance, disposables, training continuity, and infrastructure support.

Taken together, the most consequential knowledge gaps are:

- integrated control frameworks that retain transparency and stability under realistic communication imperfections,
- reproducible evaluation protocols for low-cost macro-micro systems using clinically meaningful performance metrics,
- robust, repeatable modular continuum interfaces that minimise post-swap uncertainty,
- long-horizon reliability evidence (wear, sterilisation, recalibration burden) for tendon-driven reconfigurable tools.

II.7 Priority Evidence Gaps for Additional Review

The review can be strengthened further by adding recent studies on:

- benchmark datasets and standardised test protocols for continuum surgical teleoperation,
- variable-delay and packet-loss robust bilateral/shared control in human-in-the-loop settings,
- safety assurance frameworks (fault handling, bounded motion, force-limited behaviour) for MIS robots,
- quantitative studies of modular tool-exchange repeatability and calibration drift,
- lifecycle reliability of tendon-driven instruments under repeated use and sterilisation.

III Research Question and Project Plan

III.1 Research Question and Plan Status

This section is maintained as a separate planning section that draws on the literature review but is not part of the literature-review narrative itself. At this stage, the research question and plan are presented with sufficient detail to indicate feasibility, expected resource requirements, and realistic time demands.

III.2 Plan Quality Criteria (Assessment-Aligned)

The planning section is intended to demonstrate the following progression:

- The research question and plan are clearly presented with enough detail to support feasibility and basic thesis structure.
- The plan is explicitly informed by the reviewed literature, with a clear explanation of why each major work package follows from identified evidence gaps.
- The plan includes realistic milestones, a logical thesis outline, and direct links between chapter flow, reviewed evidence, and planned validation.
- The plan includes contingencies for project variation (e.g., hardware delays, integration issues, data-quality limitations) and alternative execution pathways.

III.3 Current Planning Position

The current project plan is retained as a standalone section for subsequent development. The immediate objective is to preserve alignment with the literature-derived gaps while deferring detailed milestone refinement to the next planning pass.

IV Project Dependent Preparations

All required project-dependent preparations should be tracked here and reviewed for readiness before execution. This section should confirm whether preparations are complete and whether significant revision is still required. It should also record exceptional early progress where available, so that planning assumptions can be confirmed or adjusted and preliminary answers to research questions can be documented.

V Conclusion

The reviewed literature shows a clear convergence on three points. First, macro-micro architectures and tendon-driven continuum tools are well motivated for MIS dexterity, but remain sensitive to vibration transmission, modelling error, and teleoperation instability. Second, modern control strategies (filtering, latency compensation, predictive bilateral control, and intent-aware shared autonomy) improve performance, yet are often validated under constrained assumptions that limit direct transfer to clinically variable settings. Third, cost and adoption studies indicate that accessibility is governed by full-system burden (maintenance, disposables, training continuity, and infrastructure), not capital cost alone.

Taken together, the strongest unresolved knowledge gaps are integrated, reproducible frameworks that jointly address delay-robust control, modular reconfiguration repeatability, safety-critical reliability, and realistic cost-constrained deployment conditions.

Acknowledgements

I would like to thank...

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A Rubric-Aligned Coverage Notes (MMAN4951/MMAN9451 Interim Report)

Criterion 1: Literature Review (High Weighting)

To target Distinction/High Distinction bands, this review must do more than summarize papers. It should:

- Show conceptual links between manipulator design, control, safety, and modularity.
- Explicitly identify and justify knowledge gaps your thesis addresses.
- Include recent work (not only foundational papers).
- Critically compare methods instead of listing them.

Criterion 4: Document Presentation

For marks protection, maintain:

- consistent citation formatting,
- clear figure/table captions,
- polished grammar and structured argument flow.